

SAsp → simple As possible computer

আমরা এখানে দেখাব Ben Eaten 8 bit computer.

এখানে → everything related to Data 8 bit
→ everything " to Address 4 bit

Block diagram:

1) program counter (PC)

একটি line execute করে তার information রাখা হয় PC এ

যদি PC এ যে লাইন নিখা থাকে তা execute করে, execution এর মাধ্যমে PC ++ করে এবং পরের লাইন execute করার জন্য।

PC=1, PC++

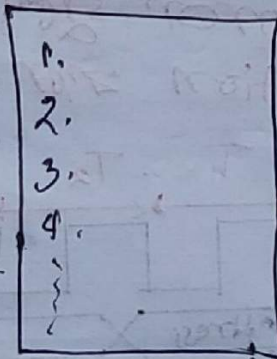
PC=2, PC++

PC=3, PC++

PC=4, PC++

...

...



1) interrupt & MAR

memory address Register

memory ka koi Location nikal kar batane ke liye
MAR (for 4 bit)

3) RAM (16x8)

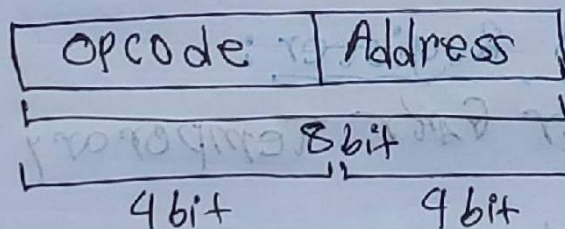
address line or
indicate karne

Each line ka size indicate karne
(8 bit)

4) Instruction Register:

Instruction Register process the instruction

instruction:



opcode :

LDA : 0000

→ Load RAM data into Accumulator

ADD : 0001

→ ADD RAM data to Accumulator

SUB : 0010

→ Subtract RAM Data from Accumulator

OUT : 1110

→ Load Accumulator Data into output register

HLT : 1111

→ Stop processing.

5) Controller sequence:

Generates conv (Traffic Police)
→ Controller sequence

6) Accumulator:

for 8 bit Register. For Arithmetic operation ~~first~~ ~~second~~ value for ~~first~~ 1st Arithmetic result is ~~first~~ 1

7) ADD/SUB

Arithmetic Logic, for 8 bit

8) B Register:

for 8 bit Temporary register, for 8 bit Arithmetic 1st and value for ~~first~~ 1st

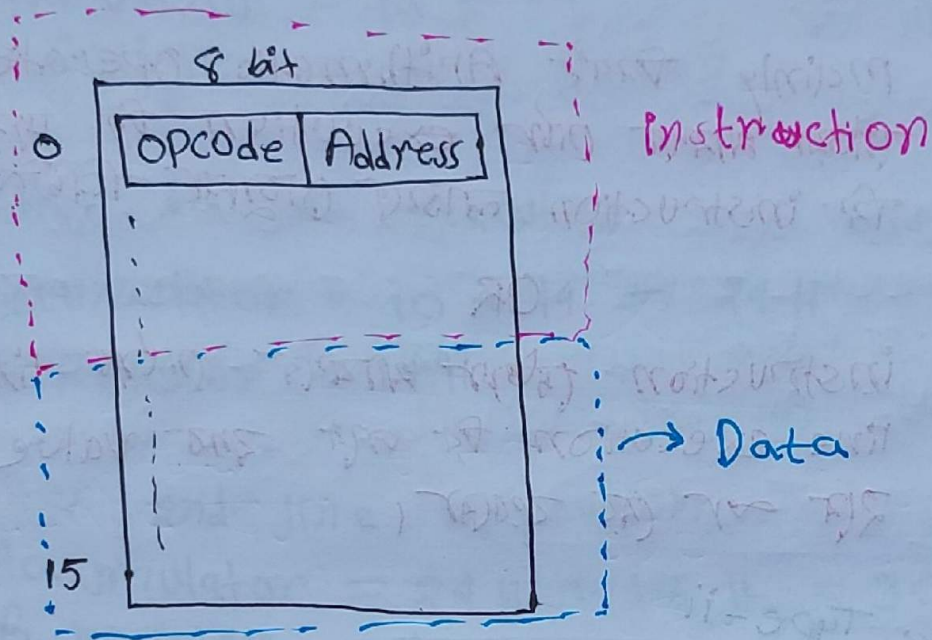
9) Output Register:

Final Output ~~first~~ 1st, 8 bit

10) Binary Display

Final result user ~~first~~ 1st

RAM



16x8 bit RAM

RAM ਵਿੱਚ ਸਾਰੇ ਆਮ ਸੇਪਰੇਟਰ ਨਹੀਂ ਹਨ।
 ਆਮ part instruction ਵਿੱਚ ਹਨ; ਆਮ part data
 ਵਿੱਚ ਨਹੀਂ ਹਨ।

instruction line by line execute ਹੋਣੀ ਚਾਹੀਦੀ ਹੈ
 ਸਫਲਤਾ ਨਾ ਮਿਲੇ HLT ਤੱਕ ਆਵੇ।

Address ਵਿੱਚ mainly RAM ਵਿੱਚ line number
 ਹੁੰਦਾ ਹੈ।

Task - 1

i) mainly for Arithmetic operation
 (উপরে শব্দটির মূল execution এর জন্য যে
 যে instruction লিখা হয়েছে সেগুলি লিখতে হবে।)

OR

ii) instruction দ্বারা শব্দটির প্রতি line by
 line execution এর জন্য যে value দি
 দেওয়া হবে সেগুলি লিখতে হবে।

Ex: Type - ii

0H	0000	1001	LDA 0H
1H	0001	1010	ADD AH
2H	0001	1011	ADD BH
3H	0010	1100	SUB CH
4H	1110	XXXX	→ OUT
5H	1111	XXXX	→ HLT
6H	XXXX	XXXX	
7H	XXXX	XXXX	
8H	XXXX	XXXX	
9H	0001	0000	→ 10H
AH	0001	0100	→ 14H
BH	0001	1000	→ 18H
CH	0010	0000	→ 20H
DH			
EH			
FH			

RAM

RAM এর ক্ষেত্রে মূলত mainy address/line
 number দ্বারা and RAM এর ডিভাইস সূচক
 একটি নির্দিষ্ট opcode দ্বারা Line number দ্বারা

After 0th line execution.

Accumulator = 10 H

B Register = X $\rightarrow$ garbage

After 1st line execution

Accumulator = 10 + 14 = 24 H

B Register = 14 H

After 2nd line execution

Accumulator = 24 H + 18 H = 3C H

B register = 18 H

After 3rd line execution

Accumulator = 3C - 20 H = 1C H

B register = 20 H

After 4th line execution

They output so Address always matter or
so don't care.

don't care register x or register value
so register value ignore register.

Process

- i) Register line for instruction for load value, then
- ii) Then Opcode (operation) register value register.
- iii) Address register value for register value
- iv) Finally operation register value register.

Caution:

RAM চিহ্ন মেমোরি লাইন Data হিচাবে চাও-
মেমোরি লাইন instruction হিচাবে চাও।

আমরা যখন মেমোরি লাইন instruction হিসাবে
execute করব তখন ২য় portion আমরা।
যখন ~~data~~ Address হিসাবে execute করব
তখন All ৪ bit will be data in Hexa

Ex

$$H\ 0E = H\ 01 + H\ 0F = \text{total memory}$$

0H	0001	0010
1H	0001	0000
2H		
...		
FH		

5th value Hexa
তৈরি করা

Explain

যখন ১st line execute করবে তখন
মেমোরি Add (0001) রাখবে 0000 line
চিহ্ন দিয়ে

so adding value ২য় 0th line চিহ্ন দিয়ে

12H (0001 0010), সিগিট (0001) Add opcode

N.B

બાદમાં HLT ના પછી આપેલા નામના PC ++
શરૂ થાય છે and તેના નામના value નીચે ।

Ex: Type - i

આમાં only ADD and SUB વાળી માટે SO
expression માં only add and sub ~~ત્રી~~ 6
correct વાળી રહે ।

જો expression નો properly execute થાય
તો આ માટે instruction સૂચના લાગે RAM ની 0th
line માં લખાઈ જાય વાળી ।

~~Remaining line સૂચના~~

expression લખાઈ એ લાગે લાગે લખાઈ ના
શરૂ instruction extra નામ લાગે ।

i) OUT

ii) HLT

Remaining line સૂચના Don't care ની
દિધા ।

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$$9 + 1 - 2 \div \frac{1}{3}$$

$$9 + 1 - 2 \times 3$$

$$9 + 1 - (2 + 2 + 2) / 9 + 1 - (3 + 3)$$

$$9 + 1 - 2 - 2 - 2 / 9 + 1 - 3 - 3$$

$$\text{Acc} = 5H$$

$$\text{B reg} = 1H$$

$$\text{Output} = 2H$$

Address	content
8H	01H
9H	48H
AH	06H
BH	03H
CH	50H
DH	02H
EH	01H
FH	04H

Data અનેર નિર્ધાર
 વિભાગ અને Answer
 સમજી મૂકી રહ
 જે ઉદાહરણ Direct
 આ પ્રકારે Answer નો
 સમજાવવાનો પ્રયત્ન કરો.

તમને Accumulator 5H already loaded
 તમને બાકી 9 સંખ્યાના પરિણામ
 4H બાકી રહેશે.

	Address	Instruction	Content
$5+4=9$ ←	0H	Add FH	0001 1111
$9+1=A$ ←	01H	Add 8H	0001 1000
$A-2=8$ ←	2H	SUB DH	0010 1101
$8-2=6$ ←	3H	SUB DH	0010 1101
$6-2=4$ ←	4H	SUB DH	0010 1101
Display ←	5H	OUT X	1101 XXXX
Stop ←	6H	HLT X	1011 XXXX
	7H	X	X X

Opcode সূত্র প্রকৃতি (দেখা
হিলে)

বিধানে B register and Output Register এর কার্যক্রম value
দিয়ে আসতে থাকে নহি, যার 5 Add/sub করার সময়
কার্য value load এর নতুন করে করে Add/sub এর।

(b)

Address	Acc	B register
0H	$5+4=9H$	4H
01H	$9+1=AH$	1H
2H	$A-2=8H$	2H
3H	$8-2=6H$	2H
4H	$6-2=4H$	2H
5H	4H	2H
6H	0	0
7H	0	0

বিস্ময় - 2 এর AT, sign
matter করে না

stop করলে সব reset
হয়ে থাকবে।

Spring 25. 04

$$(3+1) \times 2 - 5$$

$$= (3+1) + (3+1) - 5$$

$$= 3+1 + 3+1 - 5$$

Address	Content
0H	X
...	...
BH	05H
CH	12H
DH	01H
EH	06H
FH	03H

Stack Accumulate
 value value value so
 garbage load, जिसे
 लोड करके प्रशिक्षण load
 करके लाइनें

Address	Instruction
0H	LDA FH
1H	ADD DH
2H	ADD FH
3H	ADD DH
4H	SUB BH
5H	OUT X
6H	HLT X
7H	X X
...	...
AH	X X

Only Assembly code
 यानी Address and
 Instruction लिखने पर।

①

Address	content	Acc	B register
0H	0001 1111	3H	X
1H	0001 1101	$3+1=4H$	1H
2H	0001 1111	$4+3=7H$	3H
3H	0001 1101	$7+1=8H$	1H
4H	0010 1011	$8-5=3H$	5H
5H	1101 X	3H	5H
6H	1011 X	0H	10
7H	X X	X	X
:	X	X	X
:	X	X	X
AH	X X	X	X

②

Address	content	Acc	B register
0H	0001 1111	3H	X
1H	0001 1101	$3+1=4H$	1H
2H	0001 1111	$4+3=7H$	3H
3H	0001 1101	$7+1=8H$	1H
4H	0010 1011	$8-5=3H$	5H
5H	1101 X	3H	5H
6H	1011 X	0H	10
7H	X X	X	X
:	X	X	X
:	X	X	X
AH	X X	X	X

class
18th

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(a)

Address	content	Assembly code
0H	0011 110	SUB EH
1H	0100 1001	ADD 9H
2H	0001 1001	OUT X
3H	0100 1100	ADD CH
4H	1100 1101	LDA DH
5H	0011 1011	SUB BH
6H	1000 1110	HLT X
7H	0100 1010	ADD AH

(b)

Address	content	Address	Acc = 4H	Prog	out
8H	9F H	0H	4-2 = 2H	2H	3H
9H	29 H	1H	2+29 = 2BH	29H	3H
AH	07H	2H	2B+H	29H	2BH
BH	15H	3H	2B+04 = 2F	04H	2BH
CH	04H	4H	3C	09H	2BH
DH	3CH	5H	3C-15 H = 27H	15H	2BH
EH	02H	6H	0H	0H	0H
FH	1FH	7H	X	X	X

Halt program at close.

৫

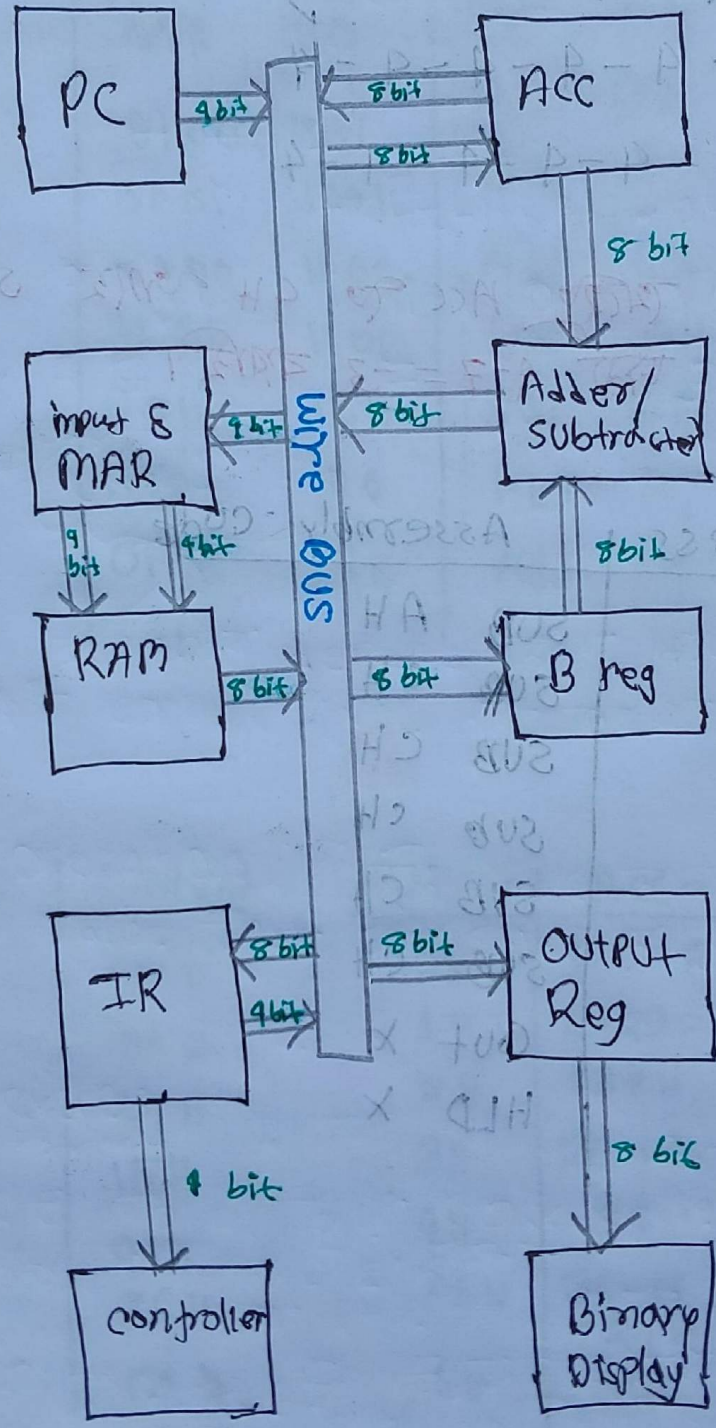
$$\begin{aligned}
 & -3 - 5 \times 4 \\
 & -3 - (4 + 4 + 4 + 4 + 4) \\
 & = -3 - 4 - 4 - 4 - 4 - 4 \\
 & = \underline{4-7} - 4 - 4 - 4 - 4 - 4
 \end{aligned}$$

যেহেতু ACC-তে ৭H আছে তাই ৫০ - ৩ বারমানের
ফল ৭-৭ = -৩ বাকি।

Address	Assembly code
0H	SUB AH
1H	SUB BH
2H	SUB CH
3H	SUB CH
4H	SUB CH
5H	SUB CH
6H	OUT X
7H	HLD X

CON

Diagram



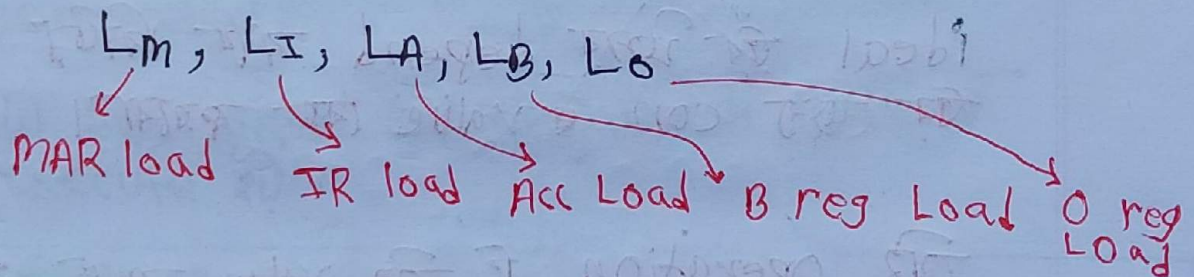
আমরা যা 808x μP এর operation সূচনা করে
 যে সূচনা Diagram এর যার যার Block use করে
 সে sequential flow এর represent করে।
 (Add, sub, LDA, HLD, ...)

State:

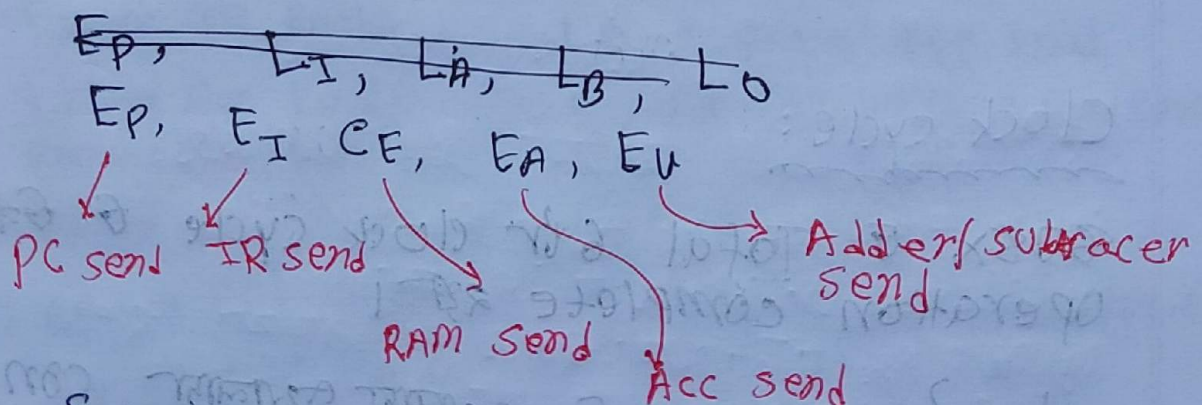
প্রতিটি Block / state এর Data দেয়া লাগে
 যা নেয়া লাগে।

কিছু কিছু Block এর 2 টাই বোনা লাগে।

i) Load (L) $\rightarrow$ incoming / receive



ii) Enable (E) $\rightarrow$ output / send



$S_u = 0 \rightarrow \text{Add}$
 $S_u = 1 \rightarrow \text{sub}$

} selector of Adder

$CP = 1 \rightarrow \text{increment pc}$
 $CP = 0 \rightarrow \text{does not " pc}$

ideal con format:

ideal के बिना यह प्रोग्राम कंप्यूटर
आमर चलाए न सकेगा वैसे ही यह प्रोग्राम भी
नहीं

$$\text{CON} = \overline{C_P} \overline{E_P} \overline{L_M} \overline{C_E} \quad \overline{L_I} \overline{E_I} \overline{L_A} \overline{E_A} \quad S_U \overline{E_U} \overline{L_B} \overline{L_O}$$

$$= 0011 \quad 1110 \quad 0011$$

$$= 3 \quad E \quad 3$$

$$= (3E3)_{16} \quad (\text{Nothing / Inactive})$$

ideal के बिना depend चलाए बिना Operation
नहीं चलाए जा सकता।

यदि Operation के लिए value चलाए न सके
तो value just swap चलाए ideal चलाए

Clock cycle:

808x के Total 6th clock cycle के बिना
Operation complete नहीं।

प्रोग्राम के T cycle के बिना चलाए जा सकता CON
generate नहीं।

আমরা প্রত্যেক instruction/operation এর জন্য
আলাদা আলাদা Diagram draw করব, এর
মিহে মিহে portion W Bus use করবে Only এর জুলা
চক্র clock cycle চ execute হবে।

- con (controller sequence) ফিরে mainly
এই Data reader or send করার মিহে
হলো হাউ
- Address related কাজ করে হলে MAR চ
মাউয়া লাহে।
- W Bus চি through হে অরহি অরহি সম্ম
connected.
- W Bus জুলা চক্র clock cycle চ
execute হবে।

pin চক্রচিতি (V.V.T)

LI → IR load

EI → IR Enable

LA → Acc Load

EA → Acc Enable

Ep → PC Enable

CE → RAM Enable

Lm → MAR Load

Lb → B reg Load

LO → Output reg Load

EU → Adder/subtractor Enable

Su = 1 → SUB

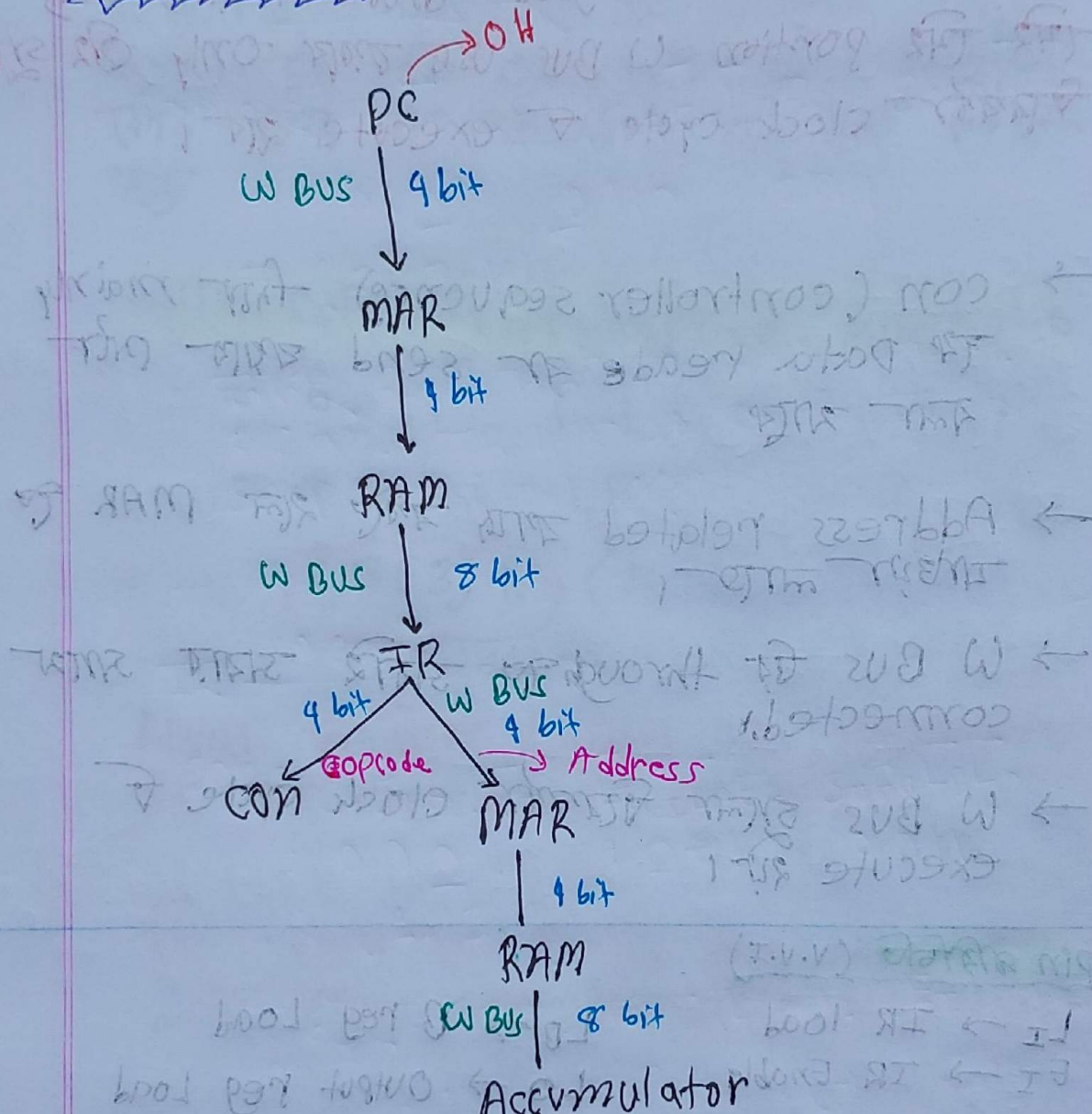
Su = 0 → Add

cp = 1 increment (PC++)

Enable → send

Load → Receive

EX LDA 9H, 9H → 10 H



For the LDA instruction full process

- i) PC to 4 bit for Address (line number 2nd)
- ii) Address to MAR to RAM
- iii) MAR to RAM to IR, RAM (for instruction fetch)
- iv) RAM to IR, Instruction Register (for data or opcode) to controller and Address to MAR

TR मॉडिफ, वगैर Address related बाफ वगैर
MAR

v) MAR ~~value~~ RAM (MAR Address निर
Value or Accumulator or गणना)

clock cycle

$T_1 \rightarrow$ Address state.

PC $\xrightarrow{\text{W Bus}}$ MAR (PC W-Bus use वगैर MAR
or address मॉडिफ)

con: $\overline{C_p} \overline{E_p} \overline{L_m} \overline{C_e} \quad \overline{L_i} \overline{E_i} \overline{L_a} E_n \quad S_u E_v \overline{L_b} \overline{L_o}$

= 0101

11110000

= (5E3 H)

चिह्न PC निर जो निर Enable (E_p) and
MAR मॉडिफ निर जो Load (L_m) swap
रह ideal or मॉडिफ

$T_2 \rightarrow$ increment state

PC ++

con: $\overline{C_p} \overline{E_p} \overline{L_m} \overline{C_e} \quad \overline{L_i} \overline{E_i} \overline{L_a} E_n \quad S_u E_v \overline{L_b} \overline{L_o}$

= 10111111

00111111

= (BE3 H)

चिह्न only C_p or use रह रह वगैर or ideal
con 2

#T₃ → memory state

RAM $\xrightarrow{W\text{ BUS}}$ IR

CON: $CpEpLm\overline{CE} \quad \overline{Ls}E_L\overline{LA}E_A \quad S_U\overline{EU}\overline{LB}\overline{LO}$

= 0010 0110 0011

= (263 H)

અહીં RAM ની સો Ram Enable (CE)
અને IR ની સો ~~IR~~ IR load (L_I)
swap કરી દેવામાં આવે છે.

#T₄

IR $\xrightarrow{W\text{ BUS}}$ MAR

CON: $CpEpLm\overline{CE} \quad \overline{Ls}E_L\overline{LA}E_A \quad S_U\overline{EU}\overline{LB}\overline{LO}$

= 0001 1010 0011

= (1A3 H)

અહીં IR સેન્ડ કરતું, Enable (E_I) અને
MAR રીસીવ કરતું Load (L_m) swap કરી
ફક્ત આદેશ આપવામાં આવે છે.

#T₅

RAM $\xrightarrow{W\text{ BUS}}$ Accumulator

CON: $CpEpLm\overline{CE} \quad \overline{Ls}E_L\overline{LA}E_A \quad S_U\overline{EU}\overline{LB}\overline{LO}$

= 0010 1100 0011

= (2C3 H)

RAM સેન્ડ કરતું સો CE અને Acc રીસીવ
સો Load (L_A) swap કરી દેવામાં આવે છે.

#T6

nothing

→ (যদিও T6 চিহ্নিত করা হয়েছে ২৫ বা ৫০, ideal con চিহ্নিত করা হবে।

CON: $C_p \bar{E}_p \bar{L}_m \bar{C}_E$ $\bar{L}_i \bar{E}_i \bar{L}_A \bar{E}_A$ $S_u \bar{E}_u \bar{L}_B \bar{L}_0$
: 0011 1110 0011
: (3E3 H)

N.B প্রকৃত con চিহ্নিত করা হবে এবং ideal চিহ্নিত করা T state চিহ্নিত করা con value চিহ্নিত করা হবে।

con value চিহ্নিত করার সময় শুধুমাত্র Binary and Hexa চিহ্নিত করা হবে।

Binary চিহ্নিত করা এবং প্রতিটি change আলাদা করে underline করা হবে।

10th class

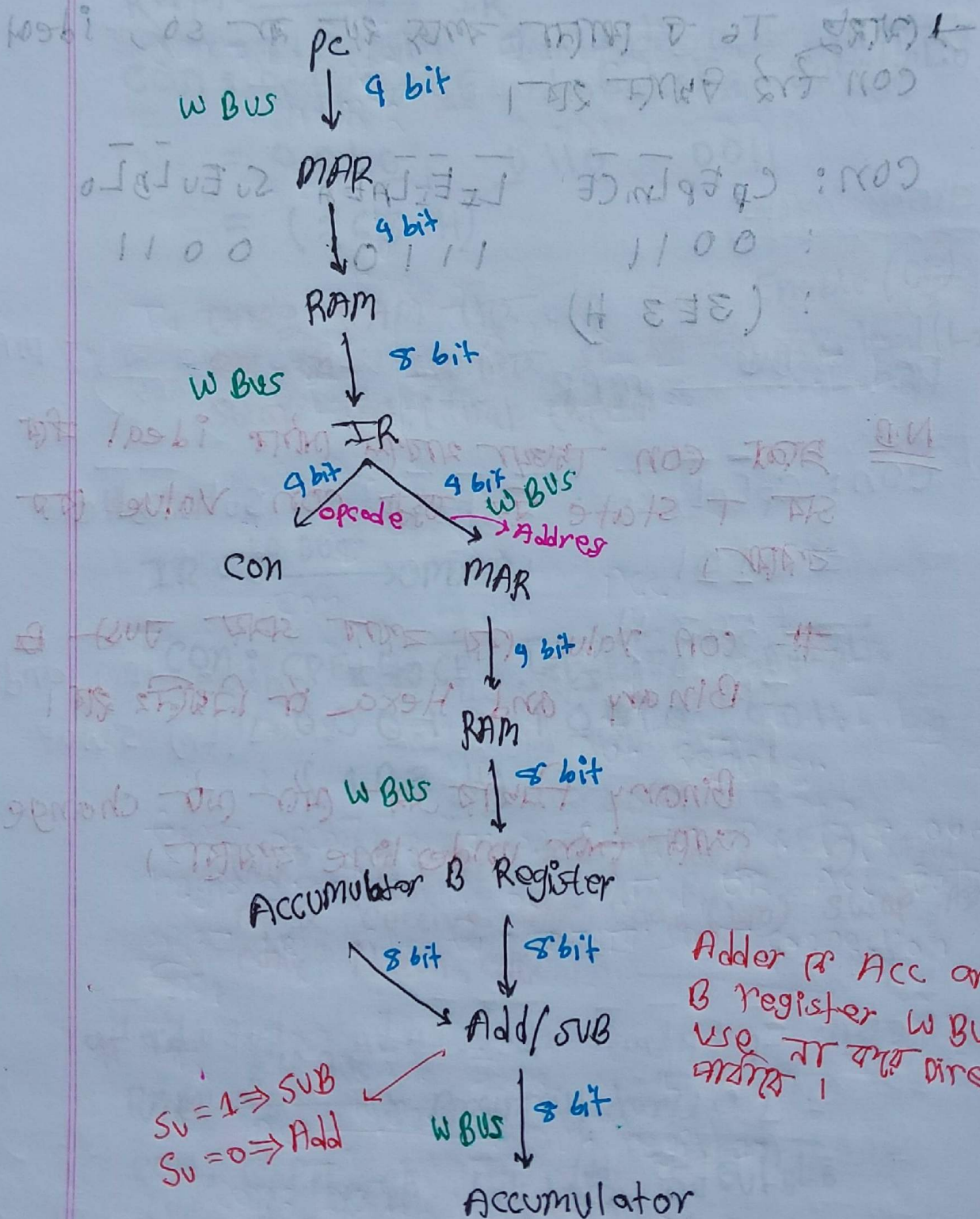
Add and sub same 2 diagram just
su for value change 2 bit

Ex: ADD AH;

ACC = 10H

BX = X

printout



$$\begin{aligned} \text{ideal con: } C_p E_p \bar{L}_m \bar{C}_E \quad \bar{L}_I \bar{E}_I \bar{L}_A E_A \quad S_V E_V \bar{L}_B \bar{L}_0 \\ = 0011 \quad 1110 \quad 0011 \\ = 3E3 \text{ H} \end{aligned}$$

$T_1 \dots T_3$ are same as LDA

$T_4 \rightarrow$ is same as LDA

$$\begin{aligned} T_5 = \text{RAM} \xrightarrow{W_{BUS}} B \text{ reg} \\ = C_p E_p \bar{L}_m \bar{C}_E \quad \bar{L}_I \bar{E}_I \bar{L}_A E_A \quad S_V E_V \bar{L}_B \bar{L}_0 \\ = 0010 \quad 1110 \quad 0001 \\ = (2E1) \text{ H} \end{aligned}$$

$T_6 \Rightarrow$ Add or SUB

= Addition/subtraction, with Adder $\xrightarrow{W_{BUS}} \text{Acc}$

= Add/SUB with 8-bit Adder, Acc to write result

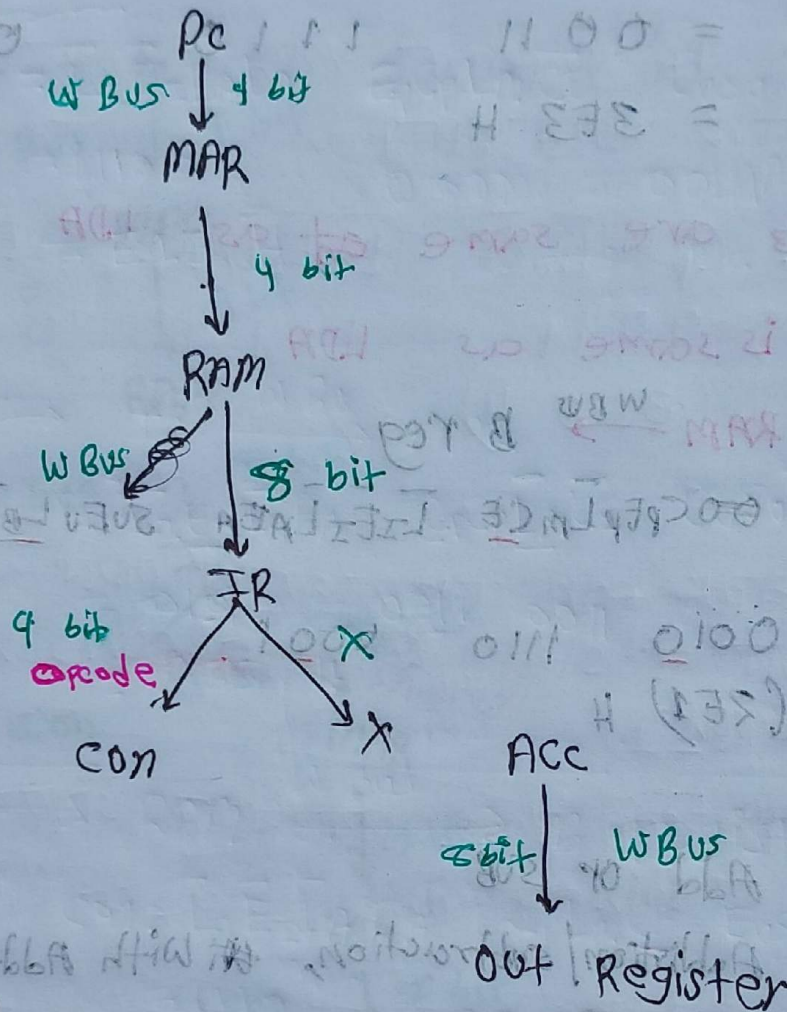
$$\begin{aligned} = C_p E_p \bar{L}_m \bar{C}_E \quad \bar{L}_I \bar{E}_I \bar{L}_A E_A \quad S_V E_V \bar{L}_B \bar{L}_0 \\ = 0011 \quad 1100 \quad 0111 \\ = (3CF) \text{ H} \end{aligned}$$

$S_V = 0$ since Add and already $S_V = 0$ BME, so don't need to flip.

if SUB

$$\begin{aligned} = 0011 \quad 1100 \quad 1111 \\ = (3CF) \text{ H} \end{aligned}$$

Ex: Out



First opcode through MAR and MAR Acc (MAR out register) Data from MAR | OR Acc to MAR Left side for register and connection.

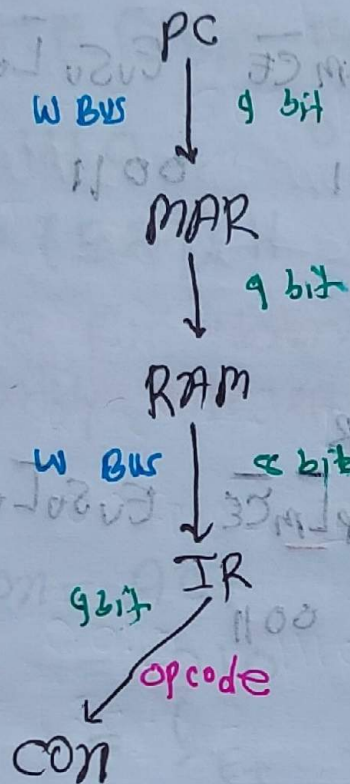
T_1, T_2, T_3 are same as LDA

T_4 Acc $\xrightarrow{W Bus}$ Out register

con: 0011 1111 0010
 $= (3F2) H$

$T_5 \rightarrow$ nothing $(3F3 H)$
 $T_6 \rightarrow$ 11 $(3F3 H)$

Ex: HLT



T_1, T_2, T_3 are same as LDA and

T_4, T_5, T_6 are disble.

Spring 25

$$\begin{aligned} \text{con} &= \bar{L}_1 \bar{E}_1 \bar{L}_A \bar{E}_A \quad E_P C_P L_m \bar{C}_E \quad E_V S_V \bar{L}_0 \bar{L}_B \\ &= 1110 \quad 0001 \quad 0011 \\ &= E13 \text{ H} \end{aligned}$$

$$\begin{aligned} T_4 &\rightarrow \text{IR} \xrightarrow{\text{W Bus}} \text{MAR} \\ &= \bar{L}_1 \bar{E}_1 \bar{L}_A \bar{E}_A \quad E_P C_P \bar{L}_m \bar{C}_E \quad E_V S_V \bar{L}_0 \bar{L}_B \\ &= 1100 \quad 0011 \quad 0011 \\ &= A33 \text{ H} \end{aligned}$$

$$\begin{aligned} T_5 &\rightarrow \text{RAM} \xrightarrow{\text{W Bus}} \text{B register} \\ \text{con: } &\bar{L}_1 \bar{E}_1 \bar{L}_A \bar{E}_A \quad E_P C_P L_m \bar{C}_E \quad E_V S_V \bar{L}_0 \bar{L}_B \\ &= 1110 \quad 0000 \quad 0010 \\ &= E02 \text{ H} \end{aligned}$$

$$\begin{aligned} T_6 &\rightarrow \text{SUB with Adder} \xrightarrow{\text{W Bus}} \text{ACC} \\ \text{con: } &\bar{L}_1 \bar{E}_1 \bar{L}_A \bar{E}_A \quad E_P C_P L_m \bar{C}_E \quad \bar{E}_V \bar{S}_V \bar{L}_0 \bar{L}_B \\ &= 1100 \quad 0001 \quad 1111 \\ &= C1F \text{ H} \end{aligned}$$

Summer -25

con: $\bar{L}_T \bar{E}_T \bar{L}_A \bar{E}_A \quad E_P \bar{C}_P \bar{L}_m \bar{C}_E \quad E_v \bar{S}_v \bar{L}_o \bar{L}_B$

= 1110 0010 0111

= (E27) H

विशेष pin observe करें operation कि 2702 or identify करेंगे।

T₄

con = A07

= 1010 0000 0111

= $\downarrow$ E_T

so, IR $\xrightarrow{W Bus}$ MAR

T₅

con = E36

= 1110 0011 0110

so, RAM $\xrightarrow[\text{LB}]{\text{CE} \quad W Bus}$ B register

T₆

con = C2F

= 1100 0010 1111

$\downarrow$ L_A

$\downarrow$ E_v

= Addition, Adder $\xrightarrow{W Bus}$ Acc

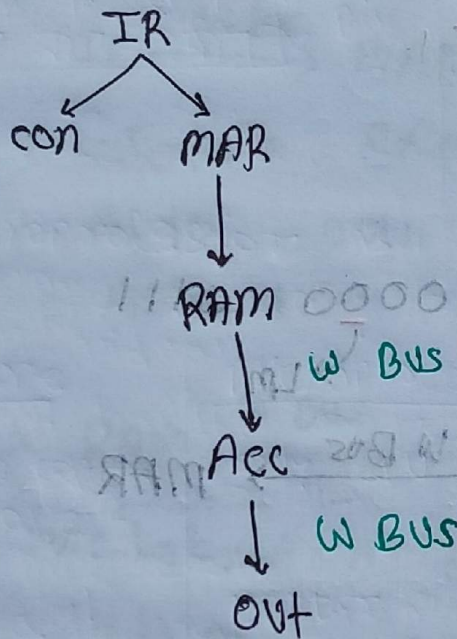
$S_v = 1 \rightarrow Add$

Normally $S_v = 0 \rightarrow Add$
 $S_v = 1 \rightarrow Add.$

Fall 25

(I) निम्न वक्तव्य

(II) निम्न वक्तव्य नूतन operation नाम LDAOUT;
LDAOUT $\rightarrow$ Load into Acc $\rightarrow$ OUT.



$$\begin{aligned}
 \text{CON} &= E_V \overline{S_V} \overline{L_B} \overline{L_B} \quad \overline{L_I} \overline{E_I} \overline{L_A} E_A \quad E_P \overline{C_P} \overline{L_M} \overline{C_E} \\
 &= 0111 \quad 1110 \quad 0010 \\
 &= (7E2)_{16}
 \end{aligned}$$

T5

$$\begin{aligned}
 \text{RAM} &\xrightarrow{\text{W BUS}} \text{Acc} \\
 &= E_V \overline{S_V} \overline{L_B} \overline{L_B} \quad \overline{L_I} \overline{E_I} \overline{L_A} E_A \quad E_P \overline{C_P} \overline{L_M} \overline{C_E} \\
 &= 0111 \quad 1100 \quad 0011 \\
 &= (7C3)_{16}
 \end{aligned}$$

T6

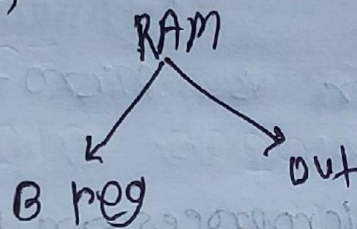
$$\begin{aligned}
 \text{Acc} &\xrightarrow{\text{W BUS}} \text{OUT} \\
 &= E_V \overline{S_V} \overline{L_B} \overline{L_B} \quad \overline{L_I} \overline{E_I} \overline{L_A} E_A \quad E_P \overline{C_P} \overline{L_M} \overline{C_E} \\
 &= 0101 \quad 1111 \quad 0010 \\
 &= (5F2)_{16}
 \end{aligned}$$

Assignment - 2

સુધુ વાંચે B અને W BUS ને જોઈને જાણે અને
EB ની પિન જાણે. Also EBxxx નામના ચાર
Hexa ને લિખીને રાખે 4 bit નામ. so EB ના સમજ
કરો don't care add વાળા.

short cut: જાણે રાખે તો B ને Load કરે then
B નામે out ને જાણે.

so,



$T_5 \Rightarrow$ RAM $\longrightarrow$ B reg and
RAM $\longrightarrow$ Output Reg

$T_6 \Rightarrow$ nothing.